

```
In [1]: import pandas as pd
```

```
In [3]: sbux = pd.read_csv("BUS303_Signature_Term_Project_Starbucks_Beverage_Menu.csv")
```

```
In [4]: sbux
```

```
Out[4]:
```

	Product_Name	Size	Milk	Whip	Serving_Size_ml	Calories	Total_Fat_g
0	Brewed Coffee - Dark Roast	Short	No Milk	No Whip	236.0	3.0	0.1
1	Brewed Coffee - Dark Roast	Tall	No Milk	No Whip	354.0	4.0	0.1
2	Brewed Coffee - Dark Roast	Grande	No Milk	No Whip	473.0	5.0	0.1
3	Brewed Coffee - Dark Roast	Venti®	No Milk	No Whip	591.0	5.0	0.1
4	Brewed Coffee Traveler - Dark Roast	1 - 236 mL serving	No Milk	No Whip	236.0	3.0	0.1
...	...	...	...	...	...	...	...
1200	White Hot Chocolate	Venti®	Whole	Whip	591.0	640.0	28.0
1201	White Hot Chocolate	Venti®	Soy	No Whip	591.0	470.0	14.0
1202	White Hot Chocolate	Venti®	Soy	Whip	591.0	540.0	21.0
1203	White Hot Chocolate	Venti®	Coconut	No Whip	591.0	450.0	17.0
1204	White Hot Chocolate	Venti®	Coconut	Whip	591.0	520.0	24.0

1205 rows × 8 columns

```
In [5]: sbux.isnull().sum()
```

```
Out[5]: Product_Name      1
        Size              1
        Milk              1
        Whip              1
        Serving_Size_ml   1
        Calories          1
        Total_Fat_g       1
        Saturated_Fat_g   2
        Trans_Fat_g       1
        Cholesterol_mg    1
        Sodium_mg         1
        Carbohydrates_g   1
        Dietary_Fiber_g   1
        Sugar_g           1
        Protein_g         1
        Vitamin_A_%DV     1
        Vitamin_C_%DV     1
        Calcium_%DV       1
        Iron_%DV          1
        Caffiene_milligrams 1
        dtype: int64
```

```
In [6]: sbux.head()
```

```
Out[6]:
```

	Product_Name	Size	Milk	Whip	Serving_Size_ml	Calories	Total_Fat_g	Saturat
0	Brewed Coffee - Dark Roast	Short	No Milk	No Whip	236.0	3.0	0.1	
1	Brewed Coffee - Dark Roast	Tall	No Milk	No Whip	354.0	4.0	0.1	
2	Brewed Coffee - Dark Roast	Grande	No Milk	No Whip	473.0	5.0	0.1	
3	Brewed Coffee - Dark Roast	Venti®	No Milk	No Whip	591.0	5.0	0.1	
4	Brewed Coffee Traveler - Dark Roast	1 - 236 mL serving	No Milk	No Whip	236.0	3.0	0.1	

```
In [7]: sbux1 = sbux.dropna(how='all')
```

```
In [8]: sbux1.isnull().sum()
```

```
Out[8]: Product_Name      0
        Size              0
        Milk              0
        Whip              0
        Serving_Size_ml   0
        Calories          0
        Total_Fat_g       0
        Saturated_Fat_g   1
        Trans_Fat_g       0
        Cholesterol_mg     0
        Sodium_mg         0
        Carbohydrates_g   0
        Dietary_Fiber_g   0
        Sugar_g           0
        Protein_g         0
        Vitamin_A_%DV     0
        Vitamin_C_%DV     0
        Calcium_%DV       0
        Iron_%DV          0
        Caffiene_milligrams 0
        dtype: int64
```

```
In [9]: sbux1.dtypes
```

```
Out[9]: Product_Name      str
        Size              str
        Milk              str
        Whip              str
        Serving_Size_ml   float64
        Calories          float64
        Total_Fat_g       float64
        Saturated_Fat_g   float64
        Trans_Fat_g       float64
        Cholesterol_mg     float64
        Sodium_mg         float64
        Carbohydrates_g   float64
        Dietary_Fiber_g   float64
        Sugar_g           str
        Protein_g         float64
        Vitamin_A_%DV     float64
        Vitamin_C_%DV     float64
        Calcium_%DV       float64
        Iron_%DV          float64
        Caffiene_milligrams float64
        dtype: object
```

```
In [13]: sbux1.iloc[1, 13] = "0"
```

```
In [11]: sbux1.head()
```

Out [11]:

	Product_Name	Size	Milk	Whip	Serving_Size_ml	Calories	Total_Fat_g	Saturat
--	--------------	------	------	------	-----------------	----------	-------------	---------

0	Brewed Coffee - Dark Roast	Short	No Milk	No Whip	236.0	3.0	0.1	
1	Brewed Coffee - Dark Roast	Tall	No Milk	No Whip	354.0	4.0	0.1	
2	Brewed Coffee - Dark Roast	Grande	No Milk	No Whip	473.0	5.0	0.1	
3	Brewed Coffee - Dark Roast	Venti®	No Milk	No Whip	591.0	5.0	0.1	
4	Brewed Coffee Traveler - Dark Roast	1 - 236 mL serving	No Milk	No Whip	236.0	3.0	0.1	

In [14]: `sbux1["Sugar_g"] = sbux1["Sugar_g"].astype(float)`

In [15]: `sbux1.dtypes`

Out [15]:

Product_Name	str
Size	str
Milk	str
Whip	str
Serving_Size_ml	float64
Calories	float64
Total_Fat_g	float64
Saturated_Fat_g	float64
Trans_Fat_g	float64
Cholesterol_mg	float64
Sodium_mg	float64
Carbohydrates_g	float64
Dietary_Fiber_g	float64
Sugar_g	float64
Protein_g	float64
Vitamin_A_%DV	float64
Vitamin_C_%DV	float64
Calcium_%DV	float64
Iron_%DV	float64
Caffiene_milligrams	float64
dtype:	object

In [16]: `sbux1.info()`

```

<class 'pandas.DataFrame'>
Index: 1204 entries, 0 to 1204
Data columns (total 20 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Product_Name                          1204 non-null   str
1   Size                                  1204 non-null   str
2   Milk                                  1204 non-null   str
3   Whip                                  1204 non-null   str
4   Serving_Size_ml                       1204 non-null   float64
5   Calories                              1204 non-null   float64
6   Total_Fat_g                           1204 non-null   float64
7   Saturated_Fat_g                       1203 non-null   float64
8   Trans_Fat_g                           1204 non-null   float64
9   Cholesterol_mg                        1204 non-null   float64
10  Sodium_mg                             1204 non-null   float64
11  Carbohydrates_g                       1204 non-null   float64
12  Dietary_Fiber_g                       1204 non-null   float64
13  Sugar_g                               1204 non-null   float64
14  Protein_g                             1204 non-null   float64
15  Vitamin_A_%DV                         1204 non-null   float64
16  Vitamin_C_%DV                         1204 non-null   float64
17  Calcium_%DV                           1204 non-null   float64
18  Iron_%DV                              1204 non-null   float64
19  Caffiene_milligrams                  1204 non-null   float64
dtypes: float64(16), str(4)
memory usage: 249.4 KB

```

```
In [31]: df2 = sbux1.dropna()
```

```
In [32]: df2.info()
```

```

<class 'pandas.DataFrame'>
Index: 1203 entries, 0 to 1204
Data columns (total 20 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Product_Name                          1203 non-null   str
1   Size                                  1203 non-null   str
2   Milk                                  1203 non-null   str
3   Whip                                  1203 non-null   str
4   Serving_Size_ml                       1203 non-null   float64
5   Calories                              1203 non-null   float64
6   Total_Fat_g                           1203 non-null   float64
7   Saturated_Fat_g                       1203 non-null   float64
8   Trans_Fat_g                           1203 non-null   float64
9   Cholesterol_mg                         1203 non-null   float64
10  Sodium_mg                             1203 non-null   float64
11  Carbohydrates_g                       1203 non-null   float64
12  Dietary_Fiber_g                       1203 non-null   float64
13  Sugar_g                               1203 non-null   float64
14  Protein_g                             1203 non-null   float64
15  Vitamin_A_%DV                         1203 non-null   float64
16  Vitamin_C_%DV                         1203 non-null   float64
17  Calcium_%DV                           1203 non-null   float64
18  Iron_%DV                              1203 non-null   float64
19  Caffiene_milligrams                   1203 non-null   float64
dtypes: float64(16), str(4)
memory usage: 249.2 KB

```

```
In [33]: sbux3 = df2.rename(columns={"Caffiene_milligrams": "Caffiene_mg"})
```

```
In [26]: sbux3.info()
```

```

<class 'pandas.DataFrame'>
Index: 1203 entries, 0 to 1204
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Product_Name          1203 non-null   str
1   Size                  1203 non-null   str
2   Milk                  1203 non-null   str
3   Whip                  1203 non-null   str
4   Serving_Size_ml       1203 non-null   float64
5   Calories              1203 non-null   float64
6   Total_Fat_g           1203 non-null   float64
7   Saturated_Fat_g       1203 non-null   float64
8   Trans_Fat_g           1203 non-null   float64
9   Cholesterol_mg        1203 non-null   float64
10  Sodium_mg             1203 non-null   float64
11  Carbohydrates_g       1203 non-null   float64
12  Dietary_Fiber_g       1203 non-null   float64
13  Sugar_g               1203 non-null   float64
14  Protein_g             1203 non-null   float64
15  Vitamin_A_%DV         1203 non-null   float64
16  Vitamin_C_%DV         1203 non-null   float64
17  Calcium_%DV           1203 non-null   float64
18  Iron_%DV              1203 non-null   float64
19  Caffeine_mg           1203 non-null   float64
dtypes: float64(16), str(4)
memory usage: 249.2 KB

```

```
In [34]: sbux3.to_csv("sbux4.csv", index=False)
```

```
In [35]: sbux4 = pd.read_csv("sbux4.csv")
```

```
In [36]: sbux4.head()
```

```
Out[36]:
```

	Product_Name	Size	Milk	Whip	Serving_Size_ml	Calories	Total_Fat_g	Saturat
0	Brewed Coffee - Dark Roast	Short	No Milk	No Whip	236.0	3.0	0.1	
1	Brewed Coffee - Dark Roast	Tall	No Milk	No Whip	354.0	4.0	0.1	
2	Brewed Coffee - Dark Roast	Grande	No Milk	No Whip	473.0	5.0	0.1	
3	Brewed Coffee - Dark Roast	Venti®	No Milk	No Whip	591.0	5.0	0.1	
4	Brewed Coffee Traveler - Dark Roast	1 - 236 mL serving	No Milk	No Whip	236.0	3.0	0.1	

```
In [37]: sbux4.equals(sbux3.reset_index(drop=True))
```

```
Out[37]: True
```

In []: